Question I is about a technologically important metallic element

Metal **A** is found in nature only in compounds, mainly as ortho silicate with the general formula $M_x(SiO_4)_y$, but also as its oxide. The oxide shows polymorphism and usually crystallizes in a monocline modification with the coordination number $CN_{metal} = 7$. At temperatures above 1100°C it will change its lattice into a tetragonal structure. Above 2000°C it shows a cubic modification. The lattice type of the latter equals the fluorite-type, in which the metallic ions form a cubic face cantered lattice with a lattice constant $a_0 = 507$ pm. The oxide anions occupy the tetrahedron gaps. This structure may be stabilized at room temperature by using CaO. The density of the pure metallic (cubic structure) oxide in question is 6.27 g/mL.

1.1. Draw the lattice components of the cubic elemental cell described into the given graphic.

1.2. What is the empirical formula of the oxide?

1.3. Which oxidation number has the metal in this oxide

1.4. The oxidation number of the metal in the oxide is the same like in the silicate. What is the empirical formula of the silicate?

1.5. What is the metal described above? Show your reasoning by a calculation.

1.6. Write down the electron configuration of the metal.

1.7. What are the coordination numbers for the cation and fort the anion?

1.8. Calculate the electron affinity $\Delta_{EA}H°$ for oxygen ($O(g) + 2\ e- \rightarrow O^{2-}(g)$) using the given thermodynamical date: $\Delta_{subl}H°(M) = 609$ kJ/mol; $\Delta_{IE}H°(M/Mn^{+}) = 7482$ kJ/mol; $\Delta_{diss}H°(O_2) = 498$ kJ/mol; $\Delta_{lattice}H°$(M-oxid) = -10945 kJ/mol; $\Delta_f H°$(M-oxid) = -1100 kJ/mol.

There is a two-step synthesis for the production of the metal. In the first step a carbo-chlorination of the silicate takes place, that means a reaction between the silicate and carbon as well as gaseous chlorine at high temperatures. Therefore, the chloride of the metal is formed (no change of the oxidation number of the metal) as well as carbon oxide and silicon tetrachloride. In a second step the metal chloride is reduced by magnesium after stripping the side products. Hydrolysing the metal chloride will give the metal oxide described above.

1.9. Write down balanced equations for the two-step synthesis of the metal.

1.10. Write down balanced equations for the hydrolysis of the metal chloride.

A series of stable halogen complexes of metal M with the coordination numbers 6, 7 and 8 is known. Let us look at a hexacoordinated complex $[MCl_2F_4]^{m+/-}$, in which the metal has the same oxidation number as in the compounds described above.

1.11. What is he charge of the complex ion? What is its name?

1.12. How many isomers are possible for this complex ion (constitution- and stereo isomers)? Draw the configuration formulae for the enantiomers.

Question II

The energy of stable states of the hydrogen atom is given by: $E_n = -2.18\times10^{-18}/n^2$ J. where n denotes the principal quantum number.

2.1 Calculate the energy differences between n = 2 (first excited state) and n = 1 (ground state) and between n = 7 and n = 1.

2.2 In what spectral range is the Lyman series lying?

2.3 Can a single photon, emitted in the first and/or sixth line of the Lyman series, ionize:

a) another hydrogen atom in its ground state?

b) a copper atom in the Cu crystal?

The electron work function of Cu is $\Phi_{Cu} = 7.44\times10^{-19}$ J.

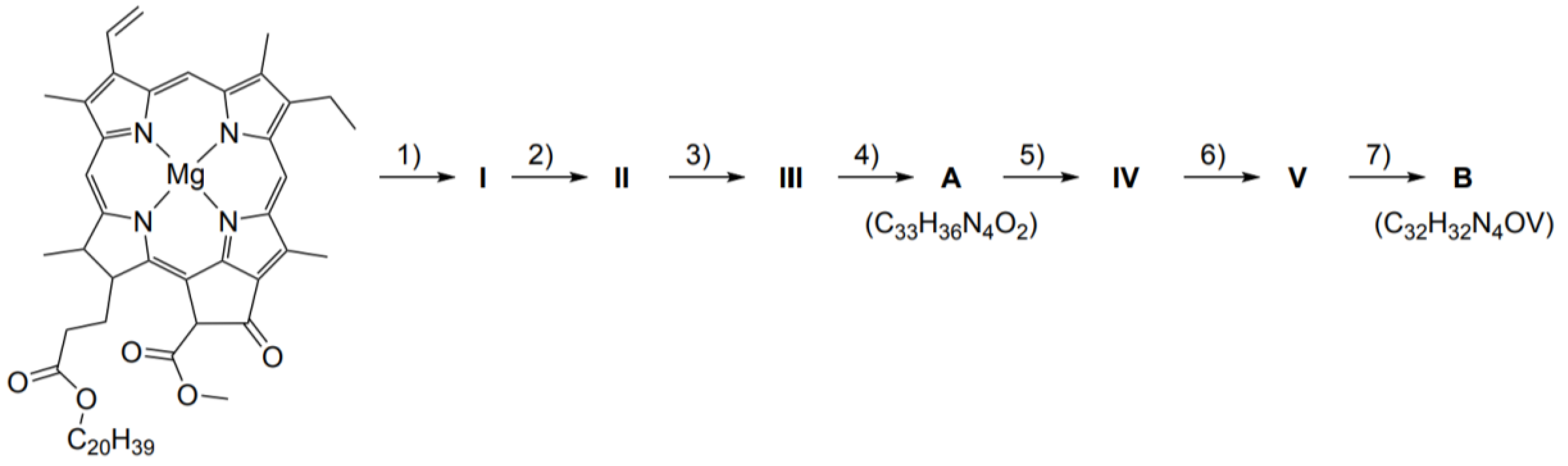
2.4 Calculate the de Broglie wavelength of the electrons emitted from a copper crystal when irradiated by photons from the first line and the sixth line of the Lyman series.

$h = 6.6256\times10^{-34}$ J s; $m_e = 9.1091\times10^{-31}$ kg; $c = 2.99792\times10^{8}$ m/s

Question 3 is about porphyrin complexes

Part A. Metals in petroleum

Saudi Arabian petroleum, primarily hydrocarbons, contains elements like vanadium in porphyrin complexes, hinting at its biological origins. Vanadium complex B is presumably formed in petroleum from chlorophyll according to the following scheme:



Step	Name
1)	Demetallization
2)	Hydrolysis
3)	Decarbomethoxylation
4)	Reduction
5)	Aromatization
6)	Decarboxylation
7)	Metal chelation

3.1 Assign the molecular formulae a–e (given in the answer sheet) to the intermediates I–V.

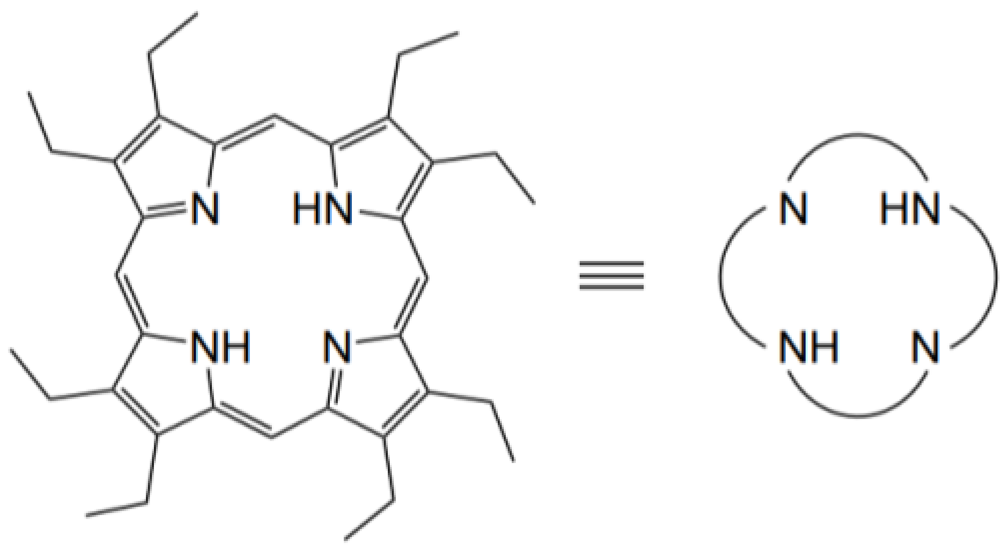
a	b	c	d	e
$C_{32}H_{34}N_4$	$C_{55}H_{74}N_4O_5$	$C_{33}H_{34}N_4O_3$	$C_{33}H_{34}N_4O_2$	$C_{35}H_{36}N_4O_5$

3.2 Draw the structures of intermediate A and vanadium complex B.

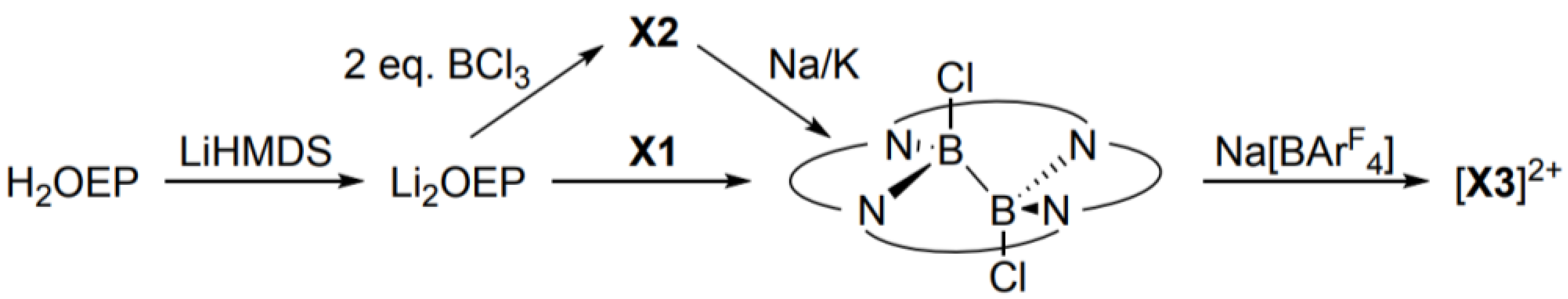
3.3 Indicate the oxidation state of vanadium in B.

Part B. Porphyrin non-metal complexes

Porphyrins are known to form chelate-type complexes not only with metals but also with some nonmetals, for example, with boron and phosphorus. Octaethylporphyrin (H_2OEP) is often used to model natural porphyrins and to study porphyrin complexes. It has the following structure and can be represented as



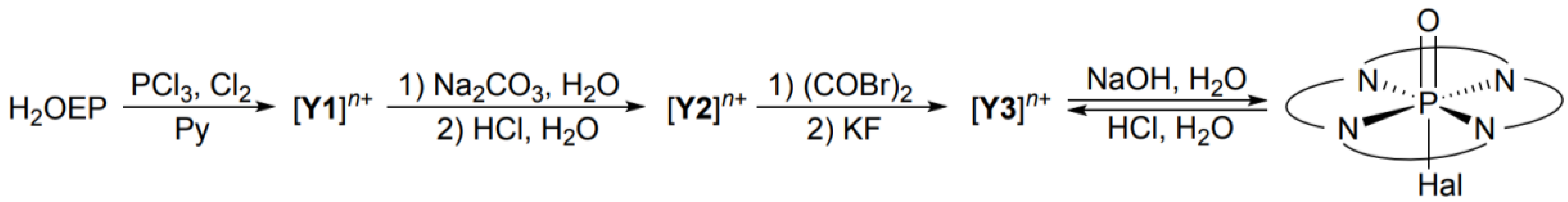
To produce dicationic planar boron porphyrin complex $[X_3]^{2+}$, either BCl_3 or another binary compound X_1 can be used as a precursor:



(LiHMDS = $LiN(SiMe_3)_2$; $Na[BAr^F_4] = Na[B\{(3,5)-(CF_3)_2C_6H_3\}_4]$)

3.4 Draw the structures of X_1, intermediate complex X_2 and the product $[X_3]^{2+}$.

Phosphorus forms porphyrin cationic complexes $[Y_1]^{n+}$, $[Y_2]^{n+}$, and $[Y_3]^{n+}$. The complex $[Y_3]^{n+}$ has one less plane of symmetry compared to $[Y_1]^{n+}$and $[Y_2]^{n+}$ (ignore ethyl substituents in OEP ligand when considering symmetry). Pyridine (Py) is used here as a basic solvent, “Hal” is one of the halogens: F, Cl, or Br:



3.5 Indicate: a) the charge "n+" of phosphorus porphyrin complexes; b) the number of planes of symmetry $N(\sigma)$ that complex $[Y_1]^{n+}$ has.

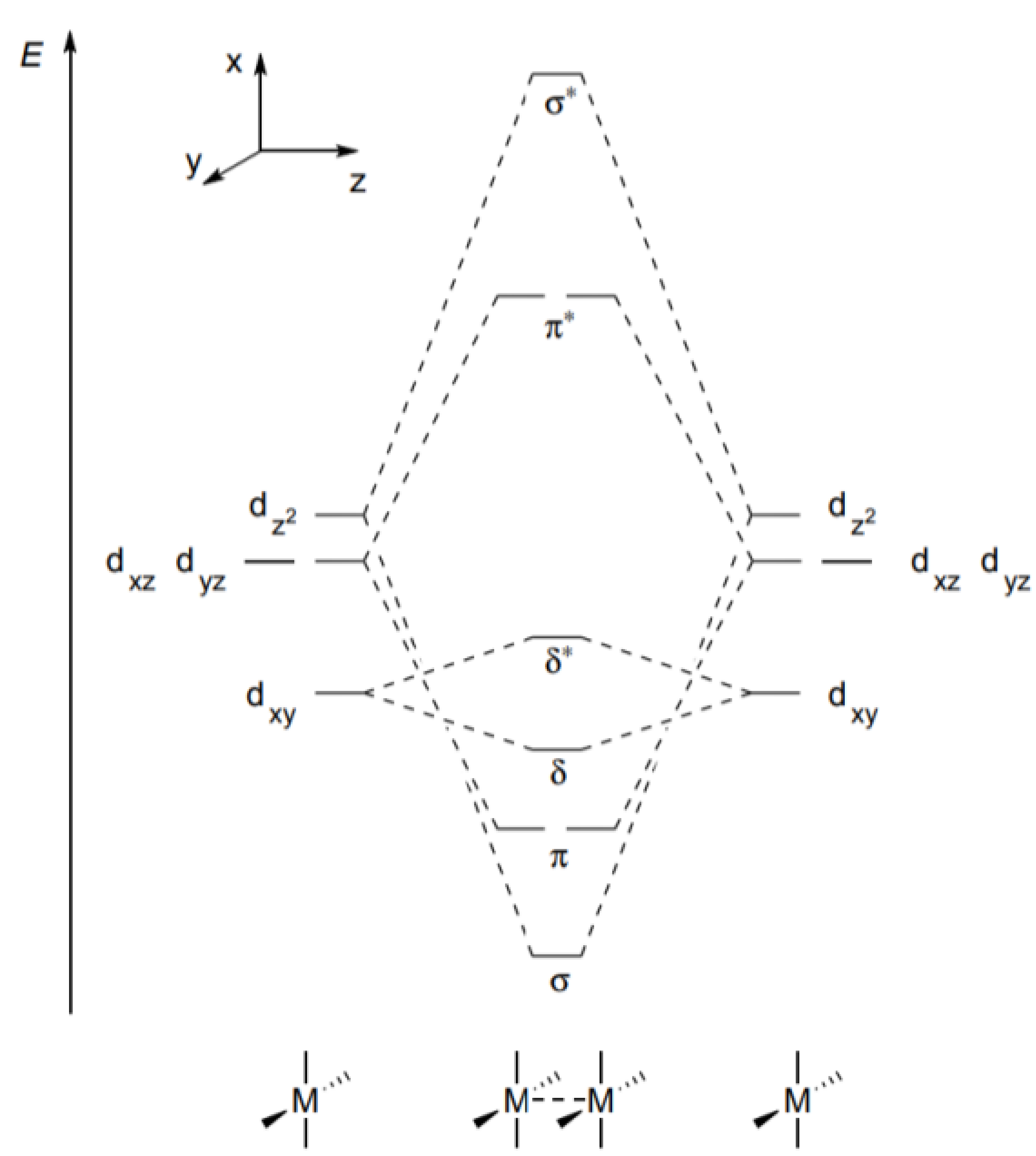
3.6 Draw the structures of $[Y_1]^{n+}$, $[Y_2]^{n+}$, and $[Y_3]^{n+}$.

Part C. Porphyrin sandwiches

Porphyrin sandwich-type complexes form when there are several porphyrin rings aligned relative to each other. Examples are $Zr(OEP)_2$, $Eu_2(OEP)_3$, $Bi_2(OEP)_2Cl_2$ that have at least 3 planes of symmetry each. Note that the hole size in the OEP ligand is ca. 4.0 Å, and the average metal–N distances in these complexes are 2.4, 2.5, 2.3 Å, respectively.

3.7 Suggest the structures of $Zr(OEP)_2$, $Eu_2(OEP)_3$, $Bi_2(OEP)_2Cl_2$. You may use a simplified representation of the OEP ligand.

Another type of porphyrin sandwich-type complex is metal–porphyrin dimers with single or multiple metal–metal bonds. For example, $[Ru(OEP)]_2$ has a Ru−Ru double bond (considering only d-electrons). Below is the molecular orbital diagram for an eclipsed dimer of the type $[M(porphyrin)]_2$ (interaction between d_{x2-y2} is not considered as these orbitals lie high in energy):



3.8 Draw the $\sigma\,(d_{z^2}+d_{z^2})$, $\pi\,(d_{xz}+d_{xz})$, $\pi\,(d_{yz}+d_{yz})$, and $\delta\,(d_{xy}+d_{xy})$ bonding orbitals according to the specified coordinate system.

3.9 Calculate the metal–metal bond order in the following complexes in eclipsed conformation: $[Mo(OEP)]_2$, $[Ir(OEP)]_2$, $[Re(OEP)]_2^+$.

The eclipsed conformation is generally preferred for metal–porphyrin dimers with 7, 8, and 9 d-lectrons. Otherwise, these complexes are more stable in a staggered conformation, where the porphyrin rings are rotated relative to each other by 45˚.

3.10 Select how the splitting (i.e., the energy between antibonding and bonding orbitals) changes for each type of orbital interaction during the conversion of eclipsed conformation into staggered:
a) increases; b) remains unchanged; c) decreases (not to zero); d) decreases to zero.

4. In the Haber–Bosch process, ammonia (NH_3) is synthesised directly from the corresponding elements using an iron-based catalyst.

1. **Write** the equation for ammonia synthesis.
2. **Determine** the temperature, T (K), of the ammonia synthesis reaction, at which $K_p = 1$, assuming $\Delta_r H°$ and $\Delta_r S°$ do not depend on temperature.

In reality, $\Delta_r H°$ and $\Delta_r S°$ depend on temperature.

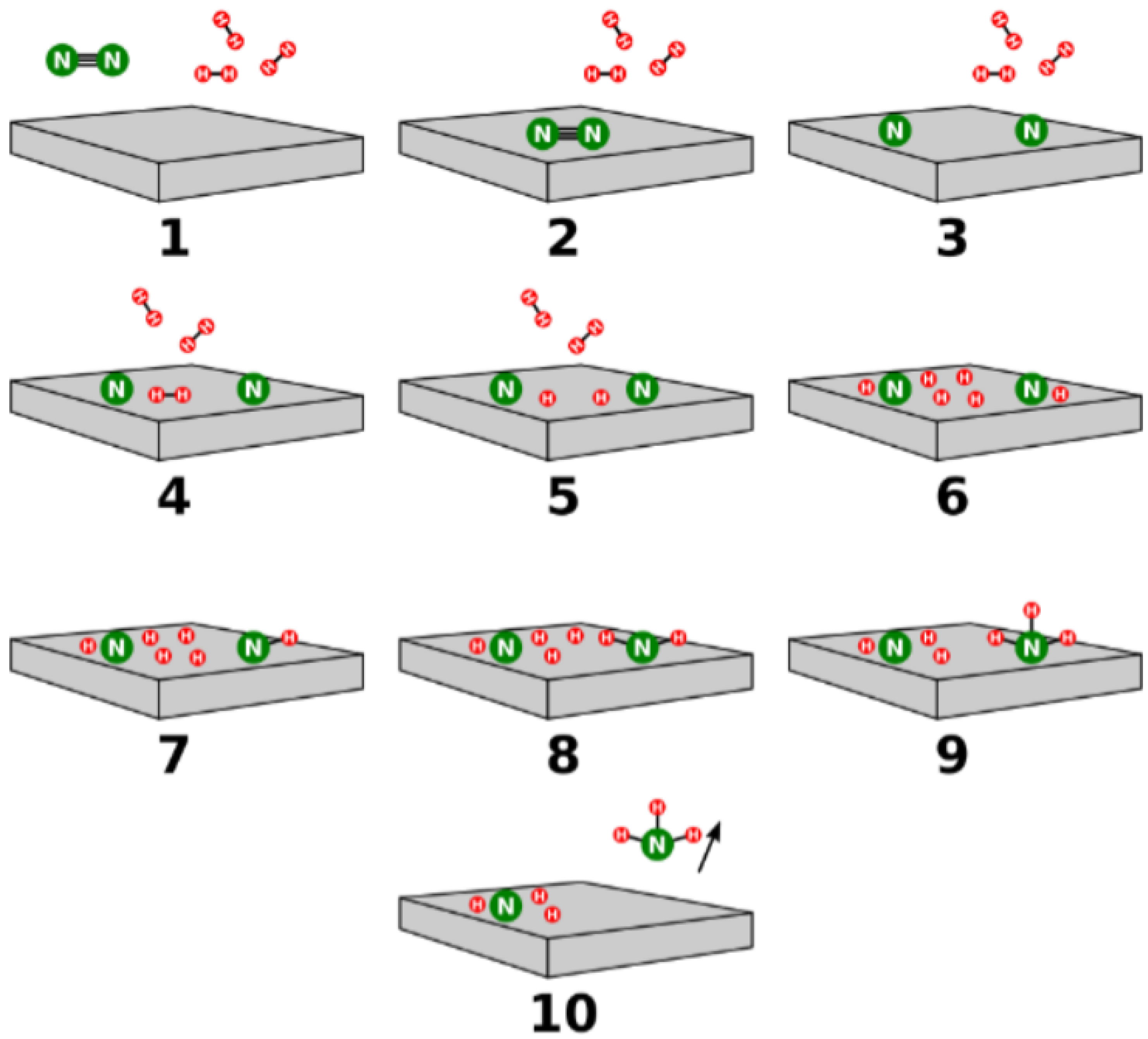
3. **Calculate** the value of K_p at 300 °C, assuming that heat capacities of the gases remain constant while enthalpy and entropy depend on temperature.
4. **Determine** the minimum amount (moles) of hydrogen required to achieve 90% conversion of 1 mol of nitrogen.

Ammonia synthesis is exothermic. Thus, low temperatures favour higher equilibrium yields, but the reaction rate decreases significantly. Industrially, the process is conducted at high temperatures and pressures to achieve optimal rate and conversion.

5. a) **Calculate** the equilibrium yield, η (%), of ammonia at 300 °C and a final total pressure of 150 bar from a stoichiometric mixture of reactants.

   b) **Explain** qualitatively how the equilibrium yield would change if the total pressure were increased while keeping the temperature constant.

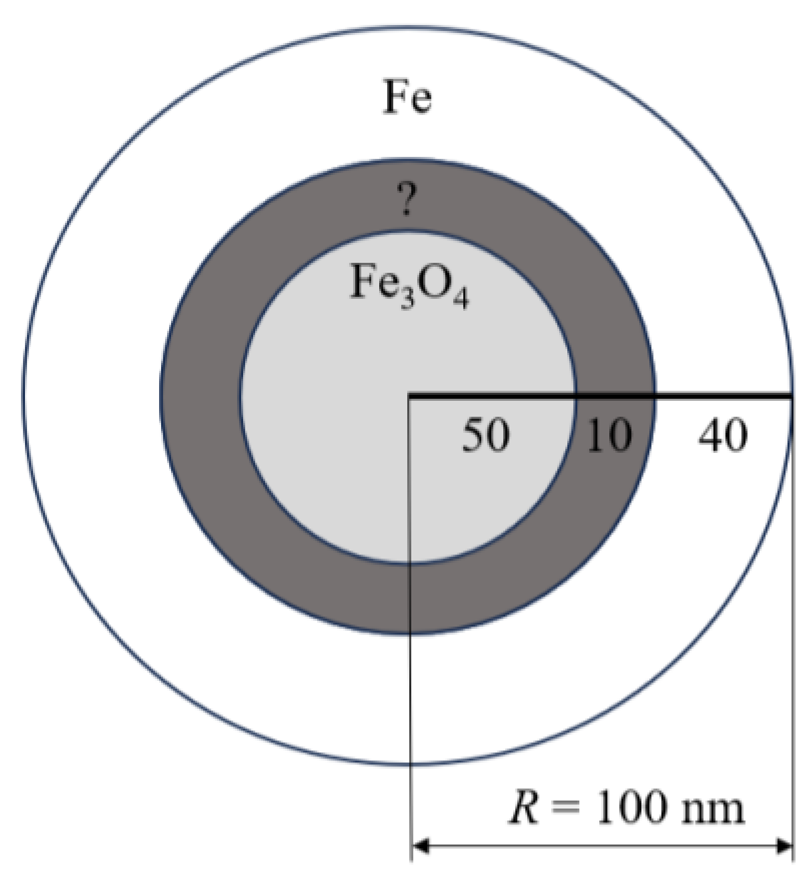
Ammonia formation proceeds via adsorption of gases on the catalyst surface, as shown in the diagram.



6. **Suggest** the possible rate-determining step of the mechanism.

The iron catalyst is produced from finely ground iron powder, which is usually obtained by reduction of high-purity magnetite (Fe_3O_4). The pulverised iron is oxidised to give magnetite particles of a specific size. The magnetite particles are then reduced. The catalyst maintains most of its bulk volume during the reduction, resulting in a highly porous high-surface-area material, which increases its catalytic efficiency.

The catalyst synthesised from 10 g of iron consists of nanoparticles with a radius, R = 100 nm, as shown in the diagram on the right.



7. **Choose** the composition of the "?" layer in the nanoparticle:

   a) Fe_2O_3 b) FeO c) FeOOH d) $Fe(OH)_3$

8. **Calculate** the total surface area of the catalyst, assuming all nanoparticles are spherical and uniform in size and composition.

9. **Calculate** how many times faster the ammonia synthesis reaction proceeds on the synthesised catalyst compared to a 10 g iron cube if the rate is directly proportional to the surface area.

Substance	$\Delta_f H°$ / kJ·mol^{-1}	$S°$ / J·mol^{-1}·K^{-1}	C_p / J·mol^{-1}·K^{-1}
H_2(g)	0	130.7	28.8
N_2(g)	0	191.6	29.1
NH_3(g)	−46.2	192.7	35.1

Substance	Fe	FeO	Fe_3O_4	Fe_2O_3	FeOOH	$Fe(OH)_3$
ρ / g·cm^{-3}	7.9	5.7	5.0	5.3	4.3	3.9